

Sample Project Documentation

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
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Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	2 Marks

Project Name

FitBuddy: AI Fitness Plan Generator

Objective

FitBuddy is an AI-assisted fitness planning application that converts a user's profile, fitness goal, and workout intensity into a personalized seven-day workout plan. It also provides a nutrition/recovery tip and allows the user to submit natural-language feedback to refine the generated plan.

Main Modules

- FastAPI web/API layer
- Pydantic input validation
- SQLAlchemy/SQLite persistence
- Gemini workout-plan generation

- Gemini nutrition/recovery-tip generation
- Feedback-based workout-plan revision
- Jinja2 web interface
- Automated tests and mock-AI fallback

Typical User Flow

1. Enter the user's name, user ID, age, weight, fitness goal, and workout intensity.
2. Generate a personalized seven-day workout plan.
3. Review the generated plan together with the nutrition/recovery tip.
4. Submit natural-language feedback to request changes.
5. View the revised plan while retaining the original plan version.
6. Inspect saved users and plans from the All Users view.

Technical Architecture

The application uses a FastAPI backend, Jinja2 templates for the web interface, Pydantic for request validation, SQLAlchemy with SQLite for persistence, and Google Gemini for AI-generated fitness content. A mock-AI/fallback path is available when the external AI service is unavailable. The API documentation is available through FastAPI's `/docs` endpoint.

Pre-requisites

1. Python 3.10+
2. FastAPI, Uvicorn and the packages listed in `requirements.txt`
3. Google Gemini API key configured through `.env` when live AI generation is required
4. Git and a GitHub account for repository management
5. A modern web browser

Project Workflow

Milestone	Activities	Where in the code
1. Project setup and validation	Configure environment, validate profile fields, prepare database and application routes	<code>app/config.py</code> , <code>app/schemas.py</code> , <code>app/database.py</code> , <code>app/routes.py</code>
2. AI generation	Build prompts and generate seven-day workout plans and nutrition/recovery tips	<code>app/gemini_generator.py</code> , <code>app/gemini_flash_generator.py</code> , <code>app/gemini_client.py</code>
3. Backend integration	Connect routes, persistence, feedback revision and result rendering	<code>app/main.py</code> , <code>app/routes.py</code> , <code>app/updated_plan.py</code>

Milestone	Activities	Where in the code
4. UI development	Build the FitBuddy form, feedback page, result page and All Users view	app/templates/, app/static/css/style.css
5. Testing and reliability	Validate routes, input handling, persistence and AI fallback behavior	tests/, tests/perf_load.py

Milestone 1: Project Setup and Workout Input

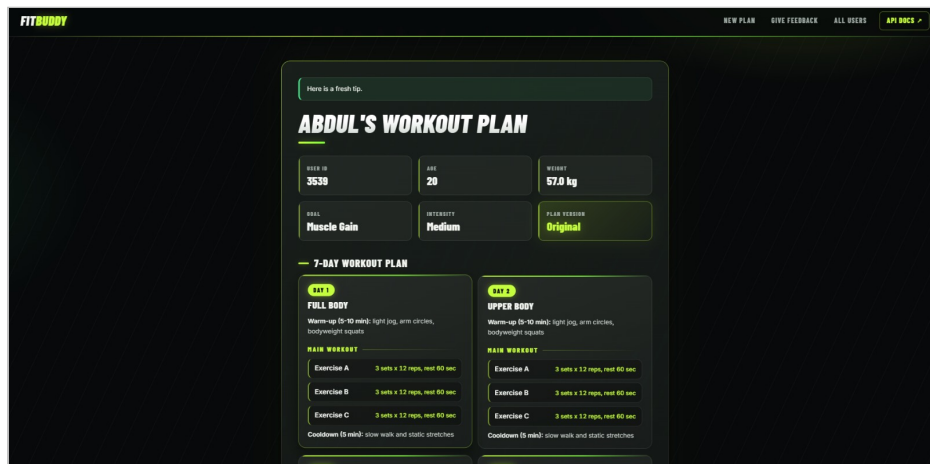
The user starts from the **New Plan** interface. The form collects the name, unique user ID, age, weight, fitness goal and workout intensity before sending the request to the backend.

New workout plan form

Figure 1 — New Workout Plan Form. The FitBuddy landing interface captures the user’s basic fitness profile and preferences before generating the seven-day plan. The User ID is used to associate feedback and saved plan information with the same user.

Milestone 2: AI-Generated Workout Plan

After successful generation, FitBuddy presents the personalized seven-day workout schedule, including the selected goal, intensity, user details, warm-up, main workout, and cooldown information.



Generated workout plan

Figure 2 — Generated Seven-Day Workout Plan. This screen demonstrates the primary AI-assisted output of FitBuddy: a structured weekly workout plan with day-wise workout content and plan metadata.

Milestone 3: Nutrition / Recovery and Feedback-Based Revision

The result interface provides a nutrition/recovery tip and a feedback form. The user can describe changes such as increasing cardio or adding more rest days, after which the application sends the feedback for plan revision.

The screenshot shows the 'NUTRITION / RECOVERY TIP' section with a 'MOCK TIP' for muscle gain: 'eat protein with every meal and drink water through the day.' Below this is a 'GET ANOTHER TIP' button. The next section is 'NOT QUITE RIGHT? TELL US WHAT TO CHANGE', which contains a 'USER ID' field (pre-filled with 3539) and a 'YOUR FEEDBACK' text area (with a placeholder example: 'e.g. More focus on cardio, include more rest days, add yoga'). A large green 'UPDATE MY PLAN' button is positioned below the feedback area. At the bottom, there are two buttons: 'CREATE ANOTHER PLAN' and 'ALL USERS'.

Nutrition and feedback section

Figure 3 — Nutrition / Recovery Tip and Feedback Revision. The screen combines a goal-aware recovery/nutrition tip with the natural-language feedback workflow used to refine an existing plan.

The dedicated feedback page provides a focused interface for entering the User ID and requested changes.

The screenshot shows a dark-themed mobile app interface. At the top, a navigation bar contains the 'FITBUDDY' logo and three tabs: 'NEW PLAN', 'GIVE FEEDBACK' (which is highlighted in green), and 'ALL USERS'. Below the navigation bar, a central card titled 'REFINE YOUR PLAN' is displayed. The card contains the following elements: a subtitle 'Enter your User ID and tell us what to change. Games will rewrite your plan.', a 'USER ID' label with a text input field containing 'The ID you used when creating your plan', a 'YOUR FEEDBACK' label with a larger text area containing the example text 'e.g. More focus on cardio, include more rest days', and a green 'UPDATE MY PLAN' button at the bottom.

Feedback form

Figure 4 — Feedback Form. The user supplies the saved User ID and describes the desired changes. FitBuddy uses this feedback to produce an updated version of the workout plan while retaining the original version.

Milestone 4: Saved Users and Plan Management

The All Users view provides a consolidated record of registered users and their plan information. It shows profile attributes, fitness goal, intensity, original plan availability, updated-plan status, feedback and a delete action.

The screenshot shows a dark-themed mobile app interface. At the top, a navigation bar contains the 'FITBUDDY' logo and three tabs: 'NEW PLAN', 'GIVE FEEDBACK', and 'ALL USERS' (which is highlighted in green). Below the navigation bar, a central card titled 'ALL USERS' is displayed. The card contains the following elements: a subtitle '3 registered users. Compare each original plan with its feedback-updated version.', a table with 10 columns (NAME, USER ID, AGE, WEIGHT, GOAL, INTENSITY, ORIGINAL PLAN, UPDATED PLAN, FEEDBACK, and an unlabeled action column), and three rows of user data.

NAME	USER ID	AGE	WEIGHT	GOAL	INTENSITY	ORIGINAL PLAN	UPDATED PLAN	FEEDBACK	
Ahmed	ahmed	25	67.0 kg	Muscle Gain	MEDIUM	View	Not updated	-	DELETE
Mike Rosta	mikerosta	29	90.0 kg	Weight Loss	HIGH	View	Not updated	-	DELETE
Isabella	isabella	19	67.0 kg	General Wellness	HIGH	View	Not updated	-	DELETE

All Users view

Figure 5 — All Users View. This screen demonstrates the application’s persistence and administrative-style view for inspecting saved FitBuddy users and their associated plan/revision state.

Milestone 5: Testing and Demonstration Evidence

The five screenshots above are actual screenshots supplied by the FitBuddy project team from the implemented application UI. They are included as visual evidence for the main workflow: plan creation, AI-generated output, nutrition/recovery guidance, feedback-based revision, and saved-user management.

The screenshots correspond to the current FitBuddy interface and are stored in `docs-src/assets/` so that the editable Markdown documentation and generated PDF documentation use the same evidence set.

Limitation

Fitness content is AI-generated general guidance and is not a substitute for medical advice, diagnosis, or professional fitness supervision.